

What is the way of finding TF-IDF of a document?

The process to find meaning of documents using TF-IDF is very similar to Bag of words,

1. Clean data / Preprocessing — Clean data (standardise data) , Normalize data(all lower case) , lemmatize data (all words to root words).
2. Tokenize words with frequency
3. Find TF for words
4. Find IDF for words
5. Vectorize vocab

Let's cover an example of 3 documents –

Document 1 It is going to rain today.

Document 2 Today I am not going outside.

Document 3 I am going to watch the season premiere.

To find TF-IDF we need to perform the steps we laid out above, let's get to it.

Step 1 Clean data and Tokenize

Word	Count
going	3
to	2
today	2
i	2
am	2
it	1
is	1
rain	1

Step 2 Find TF

Document 1—

It is going to rain today.

TF = (Number of repetitions of word in a document) / (# of words in a document)

Words/ Documents	Document 1
going	0.16
to	0.16
today	0.16
i	0
am	0
it	0.16
is	0.16
rain	0.16

TF for sentence 1

Words/ Documents	Document 1	Document 2	Document 3
going	0.16	0.16	0.12
to	0.16	0	0.12
today	0.16	0.16	0
i	0	0.16	0.12
am	0	0.16	0.12
it	0.16	0	0
is	0.16	0	0
rain	0.16	0	0

TF for the document

Step 3 Find IDF

Find IDF for documents (we do this for feature names only/ vocab words which have no stop words)

IDF = $\log[(\text{Number of documents}) / (\text{Number of documents containing the word})]$

Words	IDF Value
going	$\log(3/3)$
to	$\log(3/2)$
today	$\log(3/2)$
i	$\log(3/2)$
am	$\log(3/2)$
It	$\log(3/1)$
is	$\log(3/1)$
rain	$\log(3/1)$

IDF for document

Step 4 Build model i.e. stack all words next to each other —

Words	IDF Value		Words/ Documents	Document 1	Document 2	Document 3
going	0		going	0.16	0.16	0.12
to	0.41		to	0.16	0	0.12
today	0.41		today	0.16	0.16	0
i	0.41		i	0	0.16	0.12
am	0.41		am	0	0.16	0.12
It	1.09		it	0.16	0	0
is	1.09		is	0.16	0	0
rain	1.09		rain	0.16	0	0

Final Equation: $\text{TF-IDF} = \text{TF} * \text{IDF}$

Step 5 Compare results and use table to ask questions

Words/ Documents	going	to	today	i	am	it	is	rain
Document 1	0	0.07	0.07	0	0	0.17	0.17	0.17
Document 2	0	0	0.07	0.07	0.07	0	0	0
Document 3	0	0.05	0	0.05	0.05	0	0	0

You can easily see using this table that words like **‘it’, ‘is’, ‘rain’** are important for document 1 but not for document 2 and document 3 which means Document 1 and 2&3 are different w.r.t talking about rain.

You can also say that Document 1 and 2 talk about something **‘today’**, and document 2 and 3 discuss something about the writer because of the word **‘I’**.